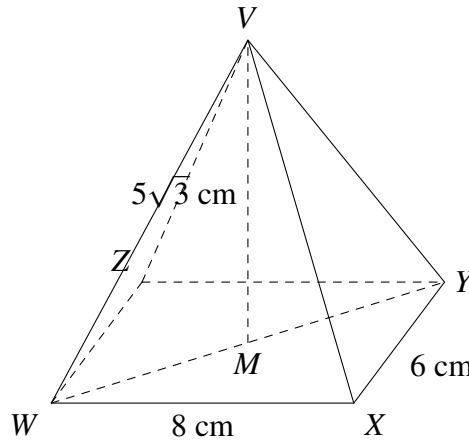


Paper 2 Practice

- 1** The diagram shows a pyramid with a rectangular base $WXYZ$.
 The apex V is directly above the centre M of the base.
 $WX = 8$ cm, $XY = 6$ cm and the vertical height $VM = 5\sqrt{3}$ cm.



- (a) Calculate the length WY .

..... [2]

- (b) Calculate the length MY .

..... [1]

- (c) Calculate the angle between the line VY and the plane $WXYZ$.

..... [2]

- 2** A cyclist rides a distance of 30 km at an average speed of x km/h. On the return journey, the cyclist increases the average speed by 5 km/h. The return journey takes 30 minutes less than the outward journey.

(a) Show that this information reduces to the equation $x^2 + 5x - 300 = 0$.

[3]

(b) Solve the equation $x^2 + 5x - 300 = 0$ to find the time taken, in hours, for the outward journey.

..... [3]

-
- 3 (a)** Find the exact value of each of the following expressions.

(i) $0.008^{\frac{1}{3}}$

..... [1]

(ii) $(-27)^{-\frac{2}{3}}$

..... [2]

(b) Simplify completely.

$$\left(\frac{2a^5}{3}\right)^3 \div 4a^7$$

..... [2]

- 4** Evaluate each of the following numerical expressions. Give your answers as integers or fractions in their simplest form.

(a) 7^0

..... [1]

(b) 4^{-2}

..... [1]

(c) 3×2^{-3}

..... [1]

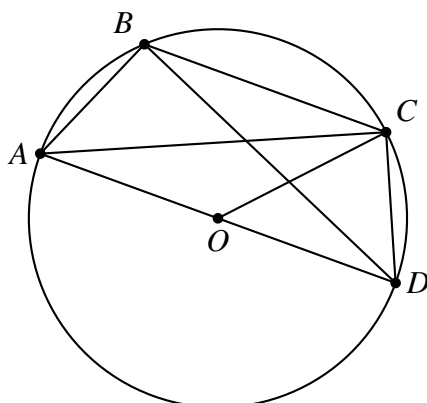
- 5** (a) A motorist drives from town A to town B at an average speed of 60 km/h, and then returns along the same route from town B to town A at an average speed of 90 km/h. Show that the average speed for the entire two-way journey is exactly 72 km/h.

[3]

- (b) The total driving time for the entire two-way journey is 3 hours and 20 minutes. Calculate the distance between town A and town B.

..... [2]

- 6 A, B, C , and D are points on a circle, centre O . The line AD is a diameter. The chord BC is parallel to AD . It is given that $\angle BAC = 24^\circ$.



NOT TO SCALE

- (a) Calculate $\angle CAD$, giving step-by-step geometric reasons.

..... [3]

(b) Calculate $\angle BCD$.

..... [2]

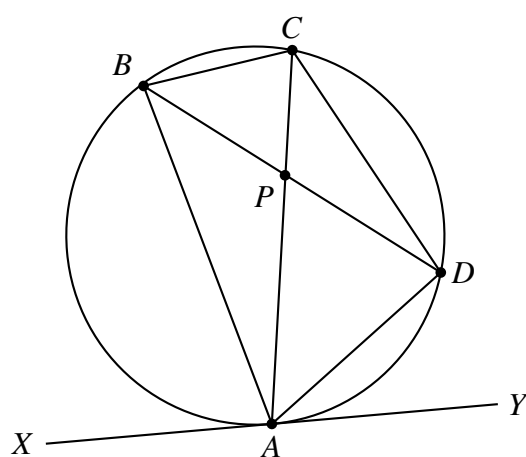
(c) Calculate $\angle COD$.

..... [1]

(d) Write down the specific geometrical name for the quadrilateral $ABCD$. Provide a reason for your answer.

..... [2]

- 7 Points A , B , C , and D lie on a circle, centre O . The chords AC and BD intersect at P . The line XY is a tangent to the circle at A . It is given that $BC = CD$, $\angle YAD = 42^\circ$, and $\angle BDC = 38^\circ$.



NOT TO SCALE

(a) Find $\angle CBD$.

..... [1]

(b) Find $\angle CAD$.

..... [1]

(c) Calculate $\angle XAB$.

..... [3]

(d) Show that $\triangle APB$ is similar to $\triangle DPC$.

[3]

-
- 8 Estimate the value of the following calculation by rounding each initial number correct to exactly 1 significant figure. Show your intermediate rounding step clearly.

$$\frac{39.4 \times 4.87}{0.193}$$

..... [3]

- 9 (a)** Expand the brackets and simplify the expression completely.

$$(3x - 2)(2x + 5)(x - 4)$$

..... [3]

- (b)** Factorise fully.

(i) $24a^3b - 54ab^3$

..... [2]

(ii) $4x^3 - 6x^2 - 16x + 24$

..... [3]

-
- 10** A company purchases a piece of industrial machinery for 50000 EUR. The value of the machinery depreciates exponentially at a rate of 20% each year.

- (a)** Show that the value of the machinery after 2 years is 32000 EUR.

[2]

- (b)** Calculate the value of the machinery after 3 years.

..... [2]

-
- 11** (a) Convert 12.5% into a fraction written in its simplest form.

..... [1]

- (b) Write the mixed number $2\frac{3}{5}$ as a decimal number.

..... [1]

-
- 12** A large cylindrical water tank has a cross-sectional area of 2.5 m^2 . Water is poured into the tank at a constant rate of 150 ml per second.

- (a) Express the flow rate of the water in litres per minute.

..... [2]

- (b) Show that the height of the water in the tank increases by 3.6 mm each hour.

[3]

- 13 (a)** Arrange the following real terms in ascending order of magnitude.

$$\frac{17}{25}, \quad 67.5\%, \quad 0.682, \quad \left(\frac{1}{8}\right)^{-1} \times 10^{-1}$$

..... [2]

- (b)** Complete the expression using the correct symbol from the set $\{>, <, \geq, \leq, =\}$.

$$3.5\% \quad \text{-----} \quad \frac{7}{200}$$

..... [1]

- 14** The mass of a breeding colony of insects increases by 10% each week. At the start of the first week, the mass is 500 grams.

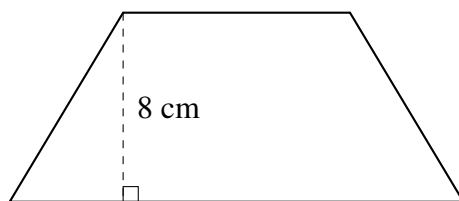
- (a)** Calculate the mass of the colony at the end of 2 weeks.

..... [3]

- (b)** Express the mass at the end of 2 weeks as a percentage of the original mass.

..... [1]

- 15** The parallel sides of a trapezium are in the ratio 3 : 5.



The perpendicular height of the trapezium is 8 cm.

The area of the trapezium is 96 cm^2 .

- (a) Calculate the lengths of the two parallel sides.

..... [3]

- (b) The trapezium is isosceles. Calculate the exact perimeter of the trapezium. Give your answer in the form $p + q\sqrt{r}$ where p, q , and r are integers.

..... [4]

- 16** (a) Evaluate the expression.

$$2 \times \sqrt{81} - 3^2 \times \sqrt[3]{27}$$

..... [2]

- (b) Evaluate the expression.

$$\sqrt{100} \times \sqrt{4} - 5^3$$

..... [2]

-
- 17** A business partnership allocates annual bonus payouts to three senior managers, A , B , and C , in a shifting ratio based on production points. Initially, £3600 is split in the ratio 3 : 4 : 5.

(a) Calculate the financial amount allocated to manager B under this initial distribution.

..... [2]

- (b) Due to an evaluation review, the distribution ratio is revised to 2 : 3 : 4 for the same total sum of £3600. Determine which manager loses out the most financially, and calculate the exact magnitude of their financial loss.

..... [3]

-
- 18** Solve the equation by factorisation.

$$3x^2 - 14x + 8 = 0$$

..... [3]

-
- 19** Evaluate the following expression, giving your final answer in standard form.

$$4.3 \times 10^5 - 7 \times 10^4$$

..... [2]

- 20** (a) Simplify completely.

$$\sqrt{32}$$

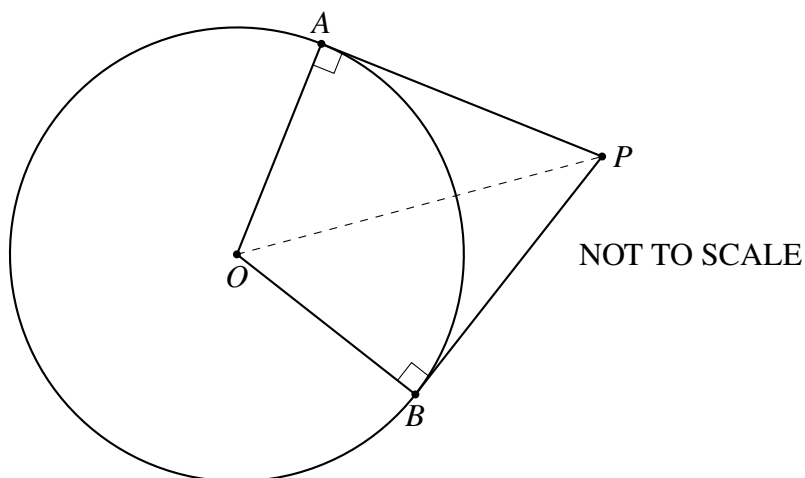
..... [1]

- (b) Rationalise the denominator and simplify.

$$\frac{12}{\sqrt{6}}$$

..... [2]

- 21** The diagram shows a circle, centre O , with radius 3 cm.
 The point P lies outside the circle such that $OP = 5$ cm.
 The lines PA and PB are tangents to the circle at the points A and B .



- (a) Show that the length of the tangent PA is 4 cm.

[2]

- (b) State the length of the tangent PB .

..... [1]

- (c) Calculate the exact area of the quadrilateral $OAPB$.

..... [2]

- 22** Work out the value of the following decimal operations without using a calculator.

(a) $3.45 + 1.8$

..... [1]

(b) 0.6×0.04

..... [1]

-
- 23** An aircraft leaves London (GMT) at 22 45 on a Tuesday.
The aircraft flies to Tokyo, which is in a time zone 9 hours ahead of GMT.
The total flight duration is 11 hours and 35 minutes.

- (a) Write the departure time of the aircraft using the 12-hour clock format, including p.m. or a.m.

..... [1]

- (b) Calculate the local time and day in Tokyo when the aircraft lands.

- (i) Local time

..... [2]

- (ii) Day

..... [1]

24 (a) Express 540 as a product of its prime factors. Give your answer in index form.

..... [2]

(b) Given that $540 \times k$ is a perfect square, where k is a positive integer, find the smallest possible value of k .

..... [2]

(c) Write down the reciprocal of k .

..... [1]